

Security Services

Introduction

- ITU-T X.800 provides the Security Architecture for OSI
- It defines a systematic way of providing security requirements

Security Services

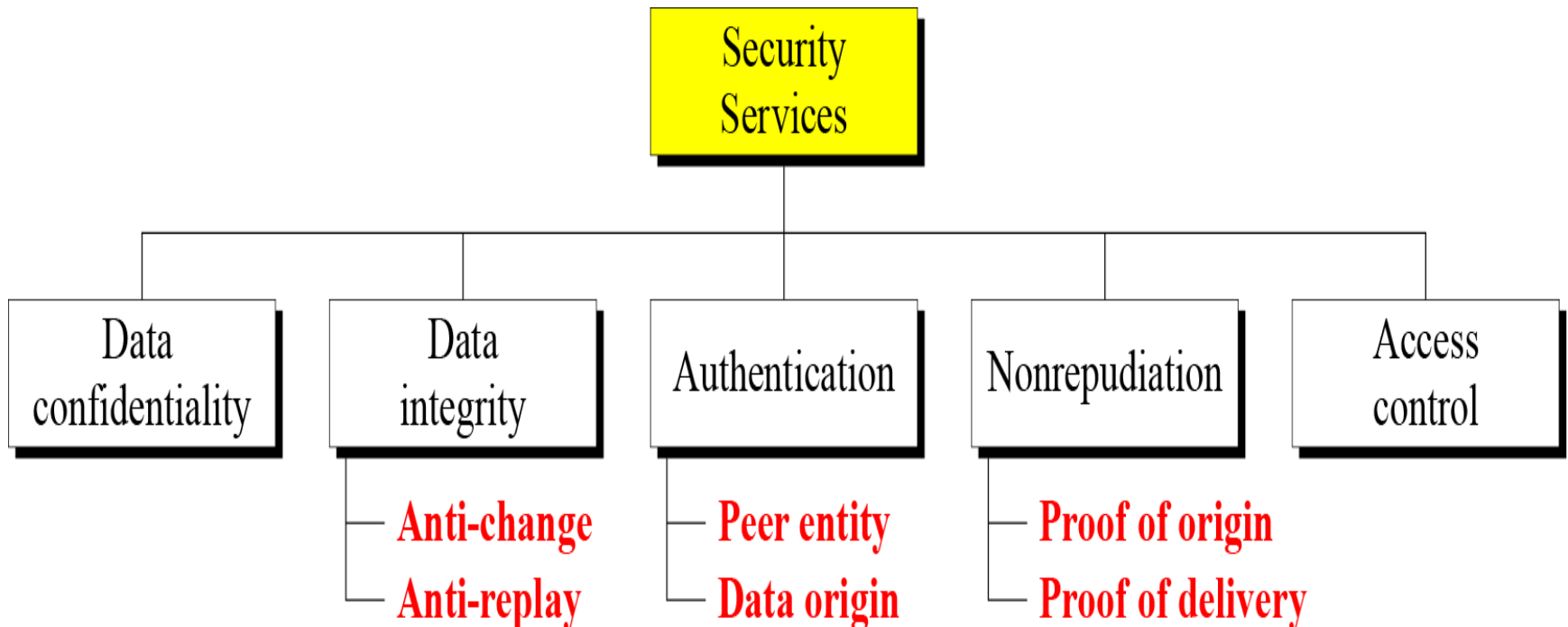
- Security Services implement security policies and are implemented by security mechanisms.
- Enhance security of data processing systems and information transfers of an organization
- Intended to counter security attacks
- Use one or more security mechanisms

Security Services

- **X.800:**
 - a service provided by a protocol layer of communicating open systems, which ensures adequate security of the systems or of data transfers
- **RFC 2828:**
 - a processing or communication service provided by a system to give a specific kind of protection to system resources

Security Services (X.800)

- X.800 divides these services into five categories and fourteen specific services.



Authentication

- Provides assurance that communicating entity is the one claimed.
 - **Peer Entity Authentication:** Used in association with a logical connection to provide confidence in the identity of the entities connected.
 - **Data Origin Authentication:** In a connectionless transfer, provides assurance that the source of received data is as claimed.

Access Control

- The prevention of unauthorized use of a resource
- This service controls
 - who can have access to a resource
 - under what conditions access can occur
 - what those accessing the resource are allowed to do

Data Confidentiality

- Assures the protection of data from unauthorized disclosure
- **Connection Confidentiality**: The protection of all user data on a connection.
- **Connectionless Confidentiality**: The protection of all user data in a single data block
- **Selective-Field Confidentiality**: The confidentiality of selected fields within the user Data on a connection or in a single data block.
- **Traffic Flow Confidentiality**: The protection of the information that might be Derived from observation of traffic flows.

Data Integrity

- Gives assurance that data received is as sent by an authorized entity.
- I. **Connection Integrity with Recovery:**
 - Provides for the integrity of all user data on a connection
 - detects any modification, insertion, deletion, or replay of any data within an entire data sequence, with recovery attempted.

II. Connection Integrity without Recovery:

- As the previous one, but provides only detection without recovery.

III. Selective-Field Connection Integrity:

- Provides for the integrity of selected fields within the user data of a data block transferred over a connection
- takes the form of determination of whether the selected fields have been modified, inserted, deleted, or replayed.

IV. Connectionless Integrity:

- Provides for the integrity of a single connectionless data block
- may take the form of detection of data modification
- Additionally, a limited form of replay detection may be provided.

V. Selective-Field Connectionless Integrity:

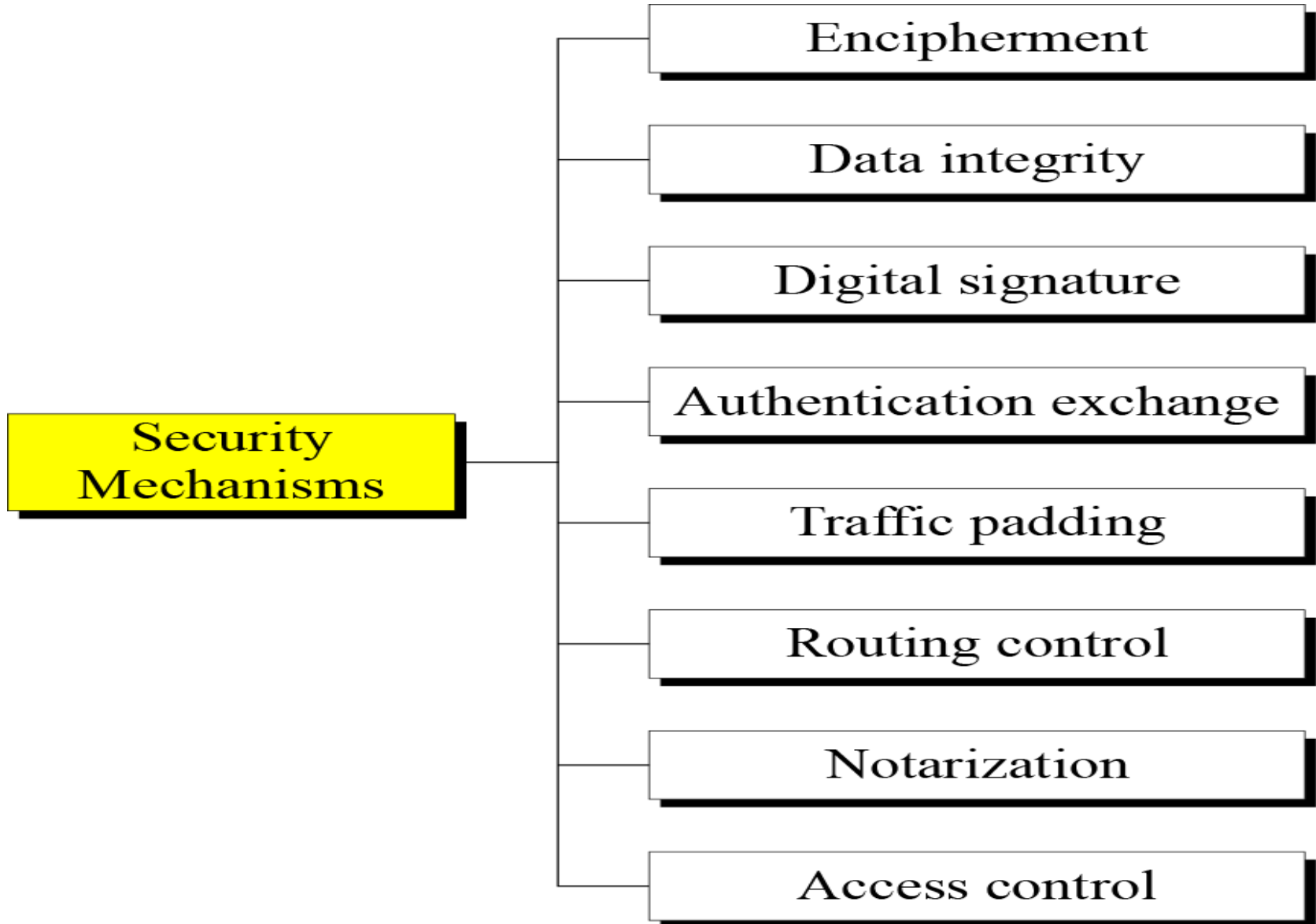
- Provides for the integrity of selected fields within a single connectionless data block
- determination of whether the selected fields have been modified.

Non-Repudiation

- It provides protection against denial by one of the parties in a communication of having participated in all or part of the communication.
 - **Non-repudiation of Origin:** Proof that the message was sent by the specified party.
 - **Non-repudiation of Destination:** Proof that the message was received by the specified party.

Security Mechanisms

- feature designed to detect, prevent, or recover from a security attack
- No single mechanism will support all services required (need several)
- However, one particular element underlies many of the security mechanisms in use:
 - cryptographic techniques



Security Mechanisms (X.800)

- **Specific security mechanisms:** –
encipherment, digital signatures, access controls, data integrity, authentication exchange, traffic padding, routing control, notarization
- **Pervasive security mechanisms:**
 - trusted functionality, security labels, event detection, security audit trails, security recovery

Security Mechanisms (X.800)

- X.800 distinguishes between reversible encipherment mechanisms and irreversible encipherment mechanisms.
- A **reversible encipherment** mechanism is simply an encryption algorithm that allows data to be encrypted and subsequently decrypted.
- **Irreversible encipherment** mechanisms include hash algorithms and message authentication codes, which are used in digital signature and message authentication applications.

Specific security mechanisms

- **Encipherment:**

- The use of mathematical algorithms to transform data into a form that is not readily intelligible.
- The transformation and subsequent recovery of the data depend on an algorithm and zero or more encryption keys.

- **Digital Signature:**

- Data appended to, or a cryptographic transformation of, a data unit that allows a recipient of the data unit to prove the source and integrity of the data unit and protect against forgery

- **Access Control:**
 - A variety of mechanisms that enforce access rights to resources.
- **Data Integrity:**
 - A variety of mechanisms used to assure the integrity of a data unit or stream of data units.
- **Authentication Exchange:**
 - A mechanism intended to ensure the identity of an entity by means of information exchange

- **Traffic Padding:**
 - The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts.
- **Routing Control:**
 - Enables selection of particular physically secure routes for certain data and allows routing changes, especially when a breach of security is suspected.
- **Notarization:**
 - The use of a trusted third party to assure certain properties of a data exchange.

Pervasive security mechanisms

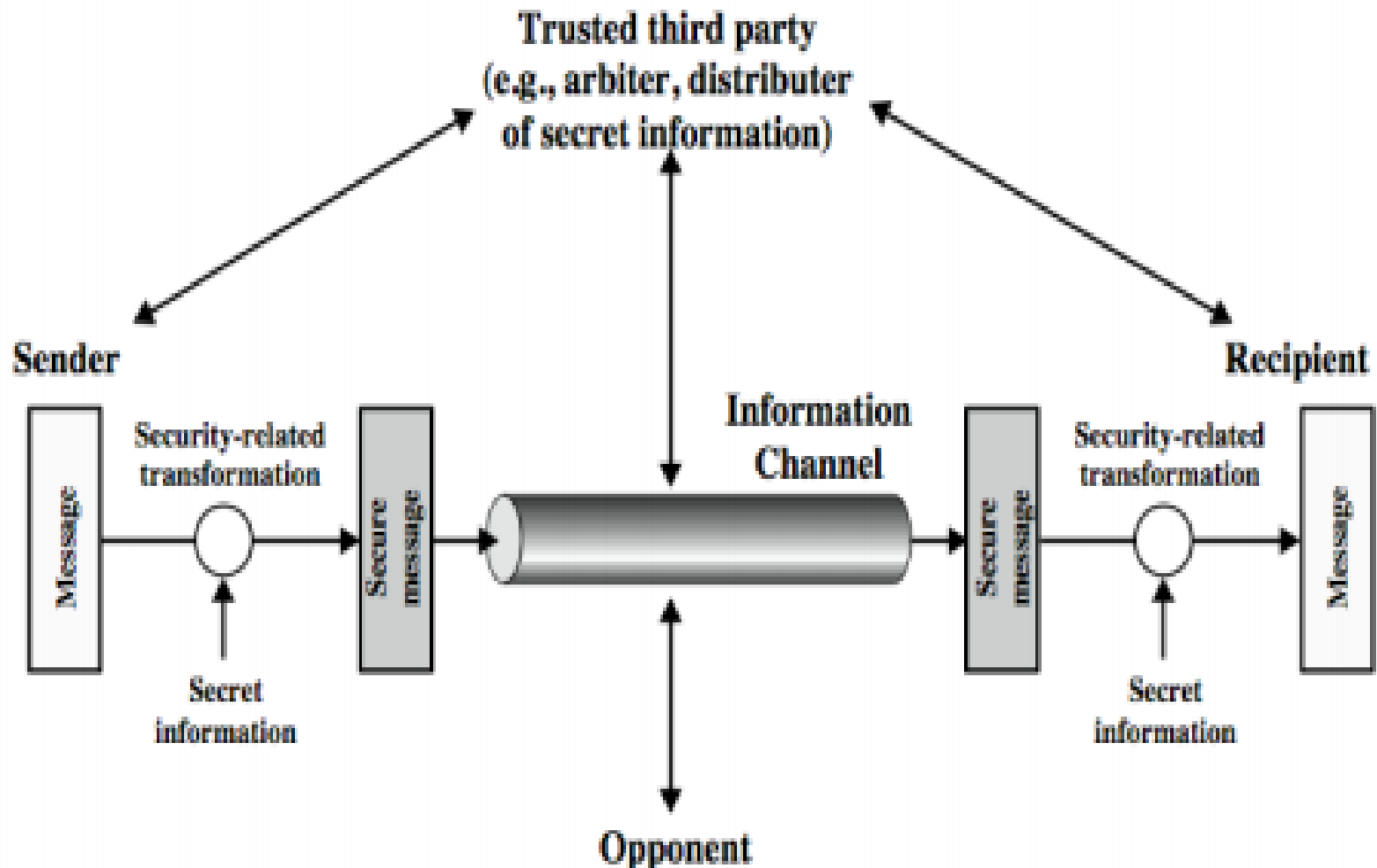
- **Trusted Functionality:**
 - That which is perceived to be correct with respect to some criteria (e.g., as established by a security policy).
- **Security Label:**
 - The marking bound to a resource (which may be a data unit) that names or designates the security attributes of that resource.
- **Event Detection:**
 - Detection of security-relevant events.

- **Security Audit Trail:**
 - Data collected and potentially used to facilitate a security audit, which is an independent review and examination of system records and activities.
- **Security Recovery:**
 - Deals with requests from mechanisms, such as event handling and management functions, and takes recovery actions.

Model of secure communication

- This model requires us to:
 - design a suitable algorithm for the security transformation
 - generate the secret information (keys) used by the algorithm.
 - develop methods to distribute and share the secret information.
 - specify a protocol enabling the principals to use the transformation and secret information for a security service

Model for Network Security



- It requires:
 - Algorithm for Security transformation
 - Secret key generation
 - Distributed and share secret information
 - Protocol for sharing secret information